Wantian Huang

Economist: empirical IO, demand estimation and pricing, causal inference and experimentation

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EDUCATION

Ph.D. Economics, University of Maryland, College Park — Expected May 2027

M.A. Economics, Columbia University — 2020

B.S. Economics & Mathematics, University of California San Diego — 2018

PROFESSIONAL EXPERIENCE

Economist Intern, Amazon Ads Marketing Finance — Seattle, WA — Jun 2026 – Aug 2026

- Evaluated five randomized promotions in an advertising-credit program and showed that credits raised seller ad adoption without incremental ad revenue, with returns varying widely across segments; estimated intent-to-treat effects by covariate-adjusted OLS, with difference-in-differences and event studies for dynamics under repeated offers.
- Found that effects fade with repeated offers and concentrate among sellers with high predicted take-up (causal forests, compliance-score matching, double machine learning), supporting capped and targeted offers.
- Authored the technical review and briefed the Ads finance director; quantified millions of dollars in annual credits with no incremental return and recommended credit-menu and re-offer changes plus a follow-on experiment.

Research Assistant to Prof. Chenyu Yang, University of Maryland — Summers 2024 – 2025

- Built and analyzed large datasets (Nielsen RMS/HMS, streaming analytics) in Python, Stata, and MATLAB; implemented econometric and structural models of demand estimation, merger simulation, and counterfactual analysis.

RESEARCH

Testing the Substitution Predictions of Demand Models: Evidence from the Blue Bell Recall

Job Market Paper

- Use the 2015 Blue Bell recall, a months-long withdrawal of a major brand, to test where its sales go; estimate logit, nested logit, BLP, and RCNL models on 3.7M product–market–months of Nielsen data with Homescan household moments.
- Find that rivals recaptured 77% of Blue Bell's volume, half via private label; non-nested models predict too little or send it to the wrong rivals, common-ρ nested models keep too much, and a nest-specific RCNL cuts prediction loss from 0.26 to 0.07.
- Show that nest structure matters even with random coefficients and that household purchase moments improve predictions; implied losses from the brand's absence differ over fourfold across models, informing assortment, stockout, and merger analysis.

Economies of Scope in Retail Distribution: A Study of Ready-to-Consume Categories

with Ying Fan, Yidan Xu, and Chenyu Yang

- Show that economies of scope raise product variety and consumer welfare, and that ignoring them understates the gains from small-firm mergers.
- Estimate demand with a BLP random-coefficients model and micro-moments on Nielsen RMS scanner data, recovering elasticities, diversion ratios, and marginal costs for merger simulation.
- Bound fixed costs under minimal assumptions using moment inequalities with optimal-transport inference; run estimation and counterfactual pipelines on an HPC cluster.

Evaluating the Welfare Impacts of EV Charging Station Subsidies: An Analysis of the NEVI Program in Texas

- Assess cost-effectiveness of 50 subsidized stations: $31.8M in fixed costs against $15.2M in annual expected welfare gains, implying roughly a two-year payback.
- Estimate EV demand with structural models that accommodate prevalent zero market shares.

Quality Uncertainty, Learning, and Product Variety in Live Streaming Platforms — *In progress*

- Quantify the welfare contribution of marginal streamers and evaluate counterfactual monetization, retention, and merger scenarios.
- Develop a nested two-stage demand framework for viewer engagement; model streamer learning, effort, and exit.

TEACHING

Instructor, University of Maryland: Economics of Regulation and Antitrust (Summer 2024)

Teaching Assistant, M.S. in Applied Economics (2021 – 2026): 29 course-terms across 11 graduate courses

TECHNICAL SKILLS

- Programming: Python (pandas, NumPy, scikit-learn, pyfixest, EconML), SQL (Redshift, Hive, Spark), Stata, MATLAB, LaTeX; Parquet panels, HPC (SLURM) and distributed computing
- Methods: A/B testing and RCTs, DiD (incl. staggered), event studies, IV/RD, causal forests, double machine learning; discrete-choice demand (logit, nested logit, BLP, RCNL), merger simulation, moment inequalities